

Antenna Datasheet

Product OC: YE0012AA
Version: 4.5
Date: 2026-08-05
Status: Released

Product Name: 4G Adhesive Mount Low Profile Monopole External Antenna

Key Features:

Frequency Band: 698–2700 MHz
Dimensions: 116.5 mm × 21.7 mm × 5.6 mm
Efficiency: Up to 66.61 %
RoHS Compliant

Overview

YE0012AA is a 4G antenna measuring 116.5 mm × 21.7 mm × 5.6 mm. This ultra-wide-band antenna provides broad coverage from 698–2700 MHz whilst offering backward-compatibility to support 3G and 2G networks as well as LTE Cat-M and narrowband IoT (NB-IoT). The antenna is available with connection via cable lengths from 250 mm, terminated with SMA Male connector. This low profile, Adhesive mount omni-directional antenna, ideal for applications where the antenna is required to be discrete, is easy to install with maximum durability assured thanks to its PC + ABS enclosure.

It allows constant and reliable transmission and reception due to its omni-directional gain across all frequency bands. The YE0012AA is designed to offers high efficiency in all working bands. It is a perfect antenna product for customers that desire highest performance. This high efficiency, high-gain omni-directional antenna is ideally suited for smart metering, remote monitoring, vehicle tracking and telematics, and many other IoT devices It is suitable for outdoor and indoor applications due to its PC enclosure.

- **Typical applications include:**

✓ Asset Tracking
✓ Smart Metering
✓ Fleet Management
✓ IoT Sensors and Modules

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

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1 Specification

Test Condition: Free Space

1.1. Electrical

Electrical	
Frequency Range	698–2700 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional

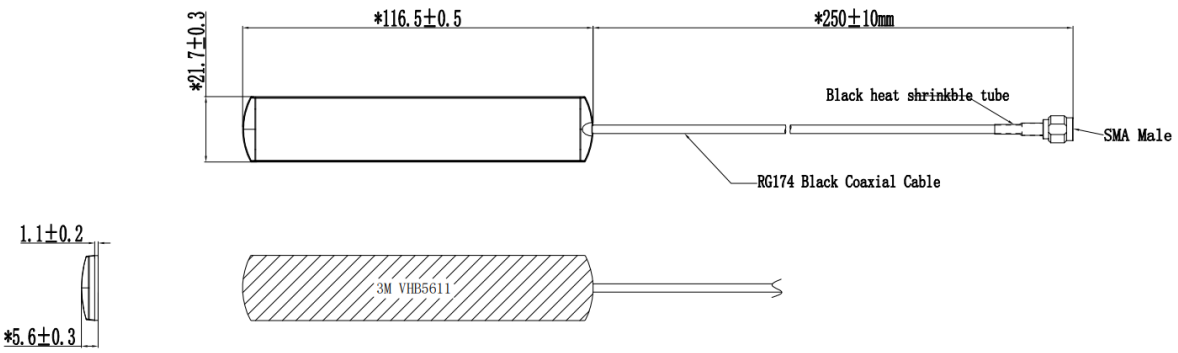
Electrical – Detail

Band / SPEC	Band	B71	B12 /B13 /B28	B5 /B8 /B26	n74 /n75 /n76	B1 /B2 /B3	B40	Wi-Fi 2G	B38 /B41	B42 /B48 /n77	n79	Wi-Fi 5G
	Freq. (MHz)	600–700	700–810	820–960	1420–1520	1700–2170	2300–2400	2400–2500	2500–2690	3300–4200	4400–5000	5150–5850
Max. VSWR		-	2.2	2.1	-	2.0	2.5	2.5	2.4	-	-	-
Max. Return Loss (dB)		-	-8.6	-9.2	-	-9.8	-7.4	-7.2	-7.7	-	-	-
AVG Eff. (%)		-	45.5	55.8	-	59.1	45.8	50.8	55.6	-	-	-
AVG. AVG Gain (dB)		-	-3.4	-2.5	-	-2.3	-3.4	-2.9	-2.6	-	-	-
Max. Peak Gain (dBi)		-	0.6	2.0	-	4.4	2.1	1.9	1.3	-	-	-
VSWR		≤ 2.5										
Return Loss		≤ -7.2 dB										
Peak Gain		≤ 4.4 dBi										

1.2. Mechanical & Environmental

Mechanical	
Antenna Dimensions	116.5 mm × 21.7 mm × 5.6 mm
Material & Color	ABS + PC & Black
Cable Type & Color & Length	RG 174 & Black & 250 mm
Connector Type	SMA Male
Mounting Type	Adhesive
Weight	Typ. 16 g
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
RoHS Compliant	Yes

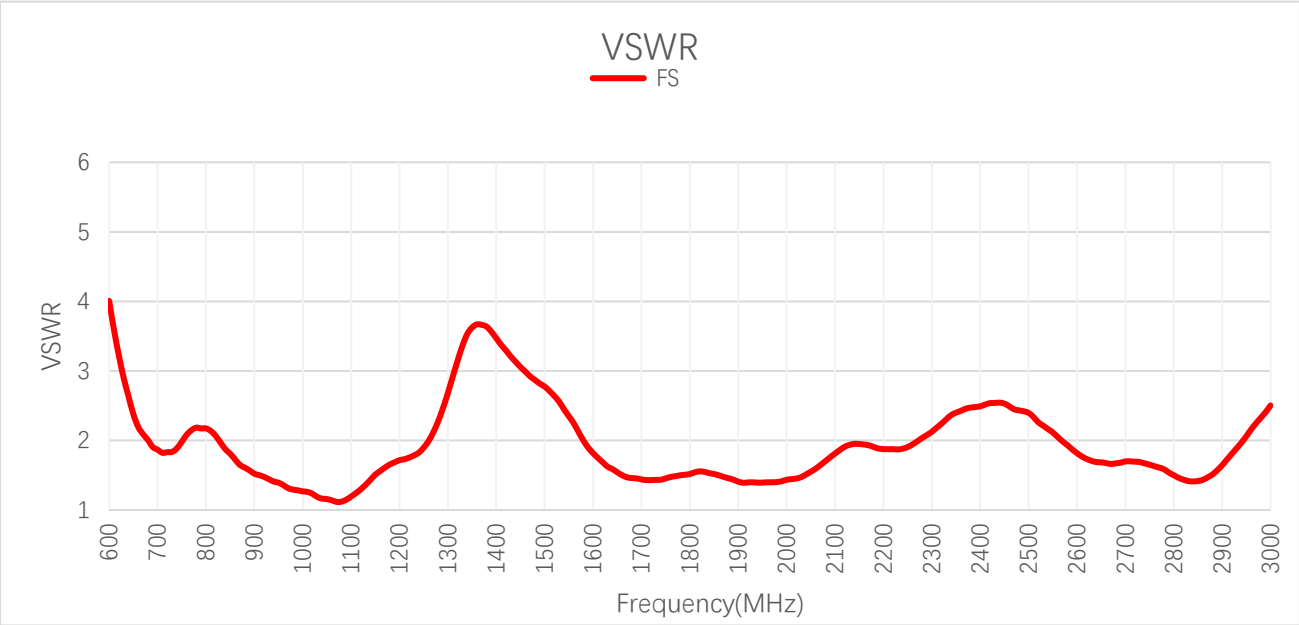
2 Drawing


*116.5±0.5
*250±10mm
*21.7±0.3
Black heat shrinkble tube
SMA Male
RG174 Black Coaxial Cable
1.1±0.2
*5.6±0.3
3M VHB5611

3 Detailed Performance

3.1. S-Parameter Test

3.1.1. VSWR



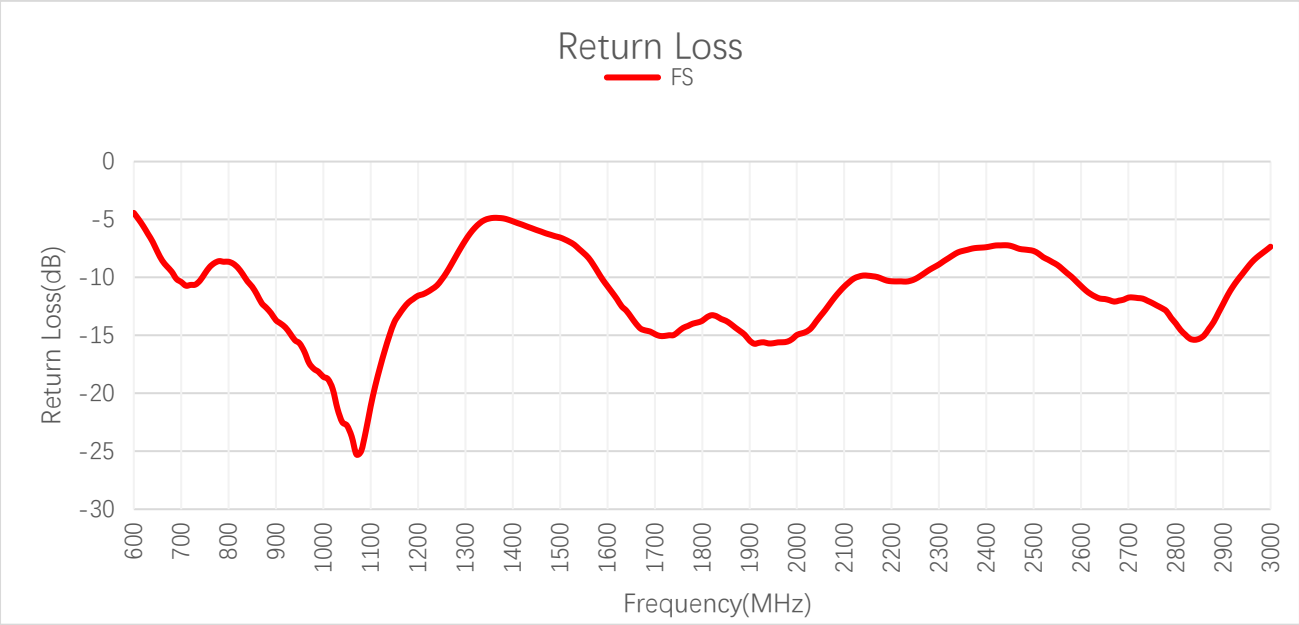
VSWR

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
VSWR	-	-	1.8	2.0	1.5	1.4	-	1.4	1.4	1.5
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
VSWR	1.4	2.0	2.4	2.5	1.8	1.7	-	-	-	-

VSWR – GNSS

Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
VSWR	-	-	-	-	-	-	2.1	-

3.1.2. Return Loss



Return Loss (dB)

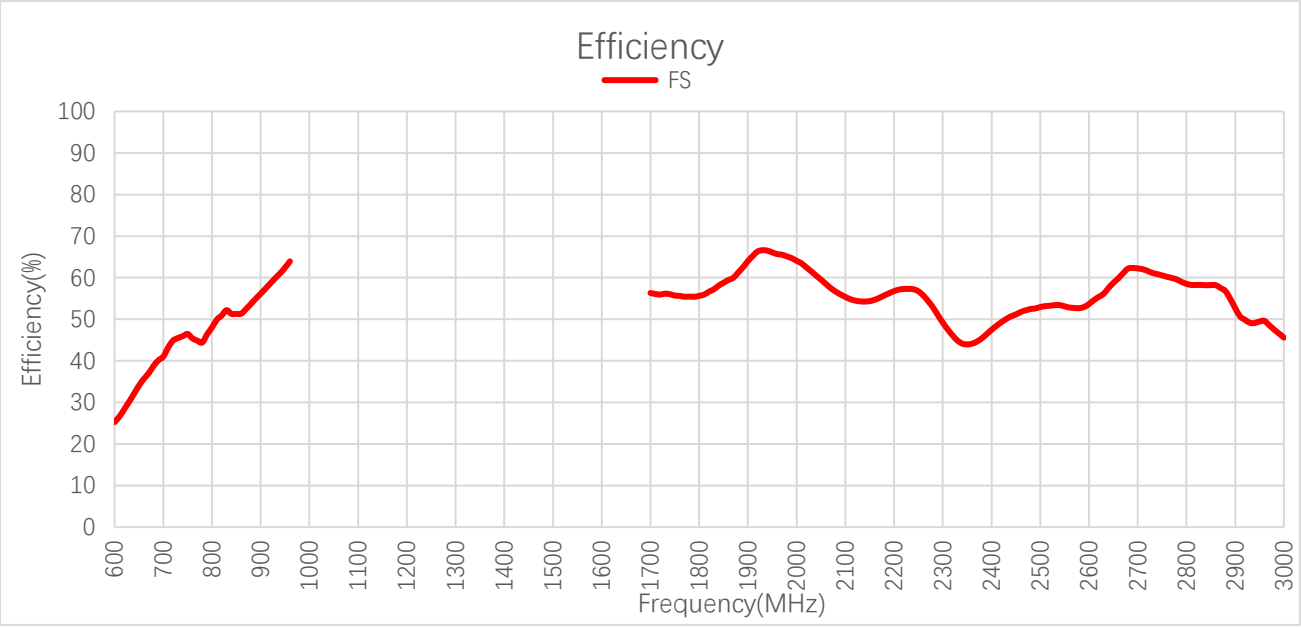
Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Return Loss (dB)	-	-	-10.7	-9.7	-13.7	-16.4	-	-15.0	-15.0	-14.6
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Return Loss (dB)	-15.7	-9.8	-7.7	-7.3	-10.7	-11.9	-	-	-	-

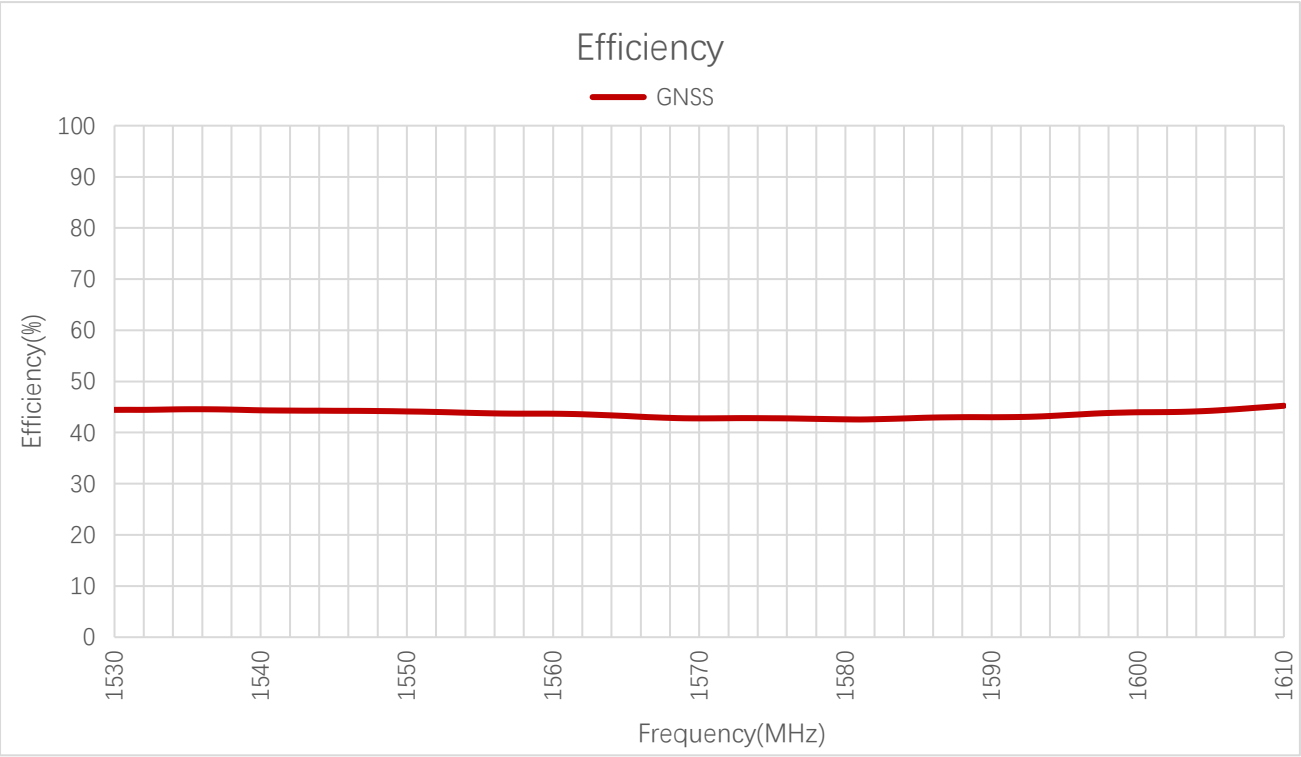
Return Loss (dB) – GNSS

Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
Return Loss (dB)	-	-	-	-	-	-	-9.0	-

3.2. Radiation Performance Test

3.2.1. Efficiency





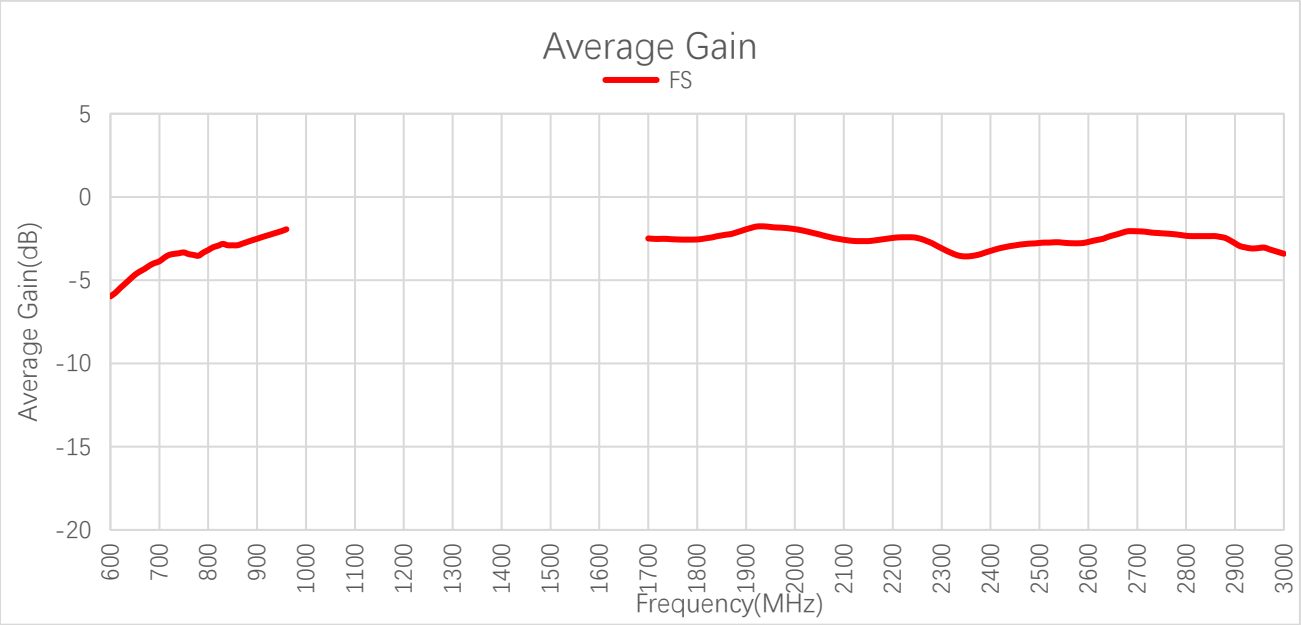
Efficiency (%)

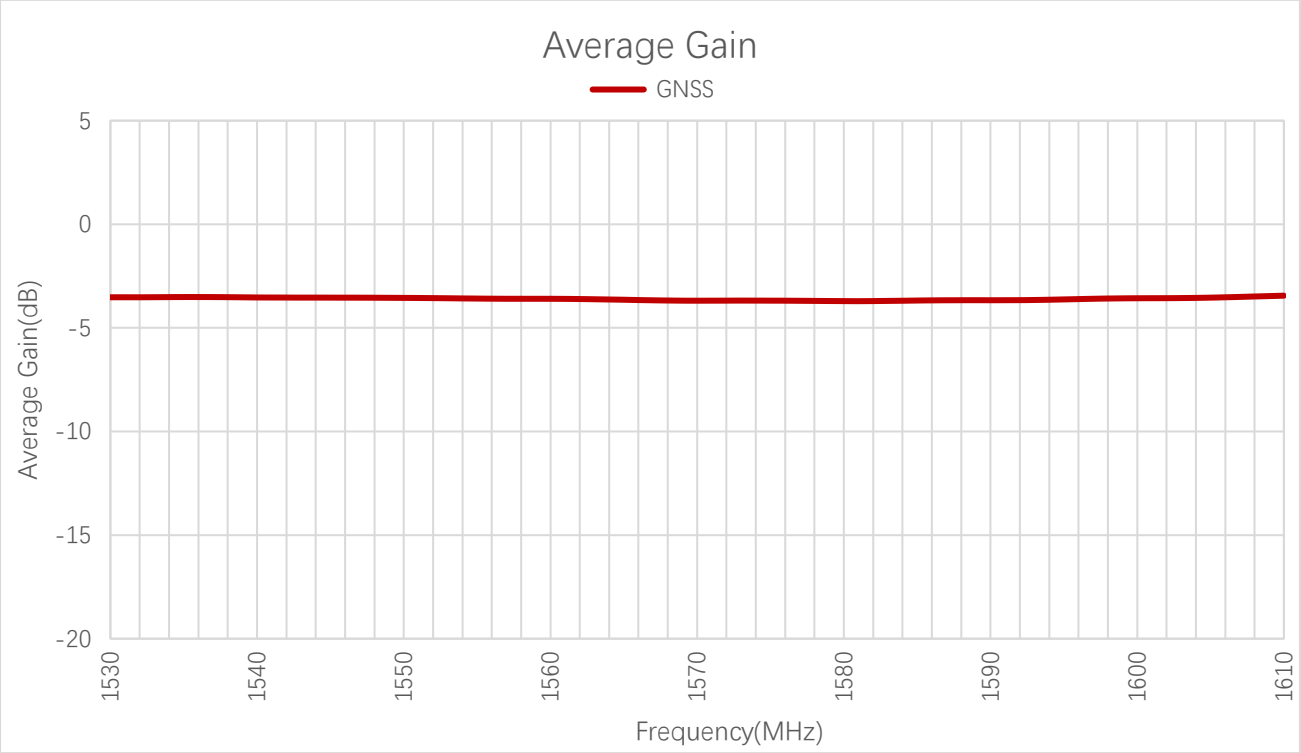
Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Efficiency (%)	-	-	43.2	52.2	56.2	63.9	-	56.1	56.0	61.3
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Efficiency (%)	66.1	54.3	44.0	51.2	53.7	62.3	-	-	-	-

Efficiency (%) – GNSS

Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
Efficiency (%)	-	-	-	-	-	-	42.8	-

3.2.2. Average Gain





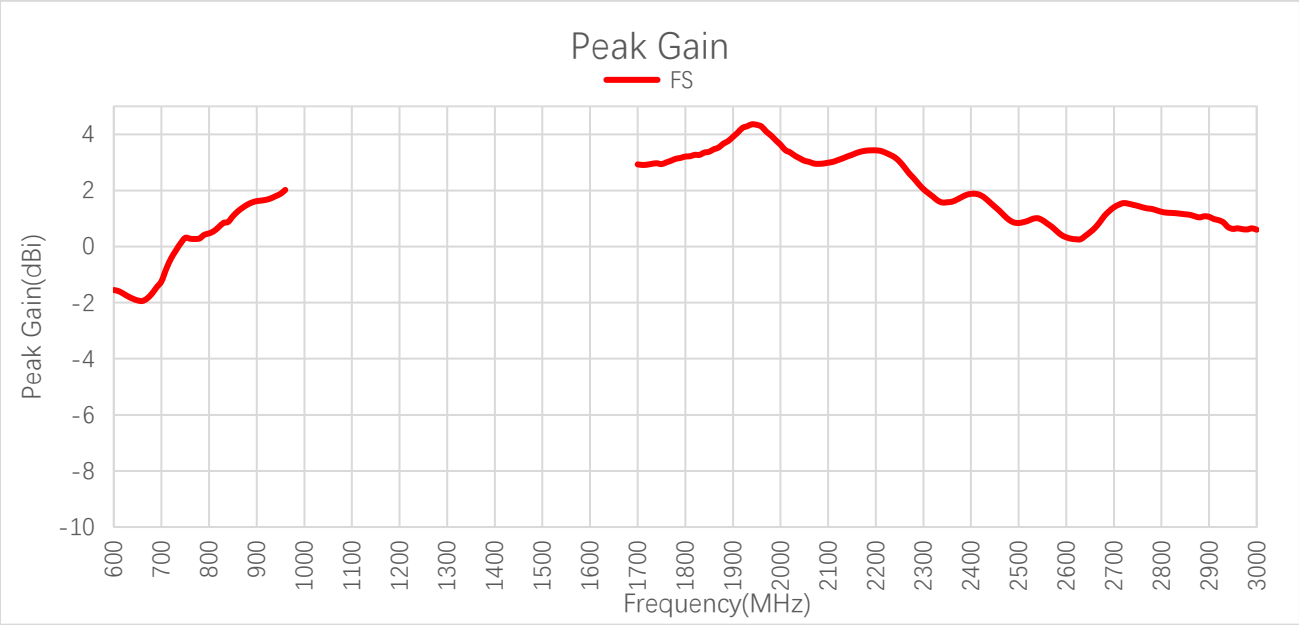
Average Gain (dB)

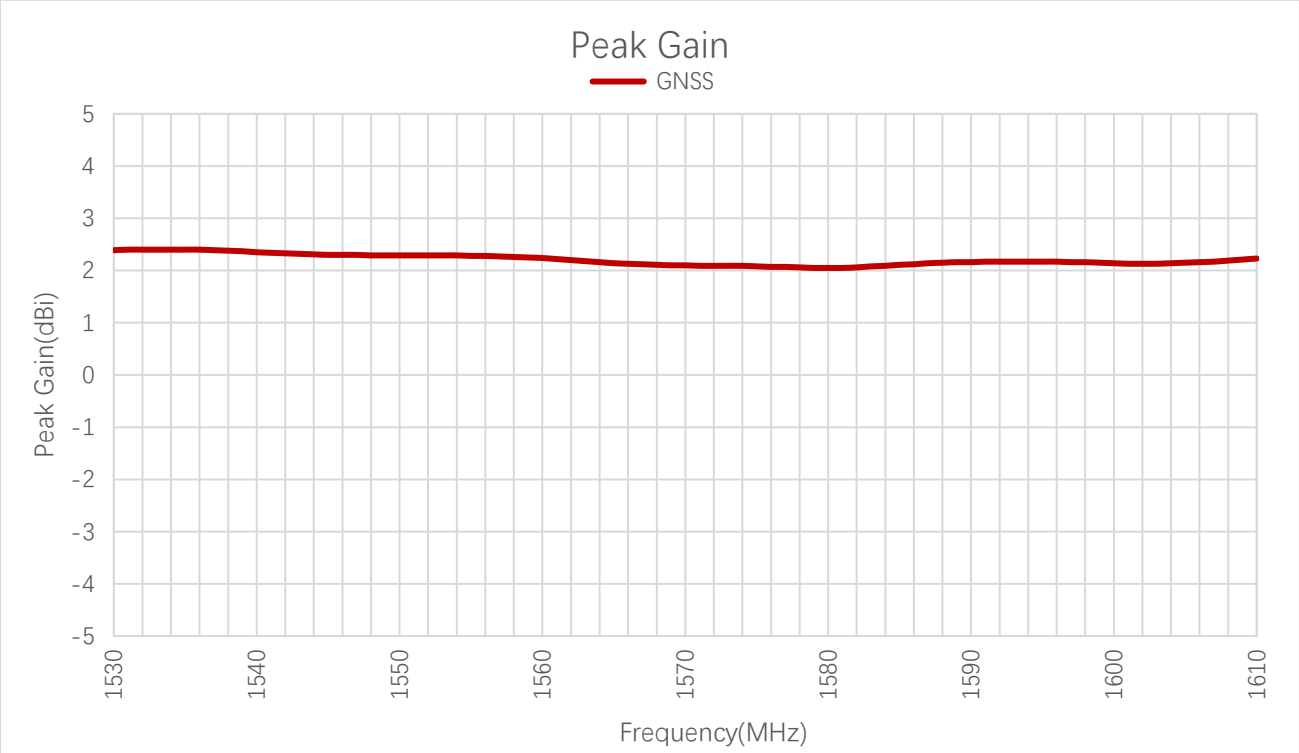
Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Average Gain (dB)	-	-	-3.7	-2.8	-2.5	-1.9	-	-2.5	-2.5	-2.1
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Average Gain (dB)	-1.8	-2.7	-3.6	-2.9	-2.7	-2.1	-	-	-	-

Average Gain (dB) – GNSS

Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
Average Gain (dB)	-	-	-	-	-	-	-3.7	-

3.2.3. Peak Gain





Peak Gain (dBi)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Peak Gain (dBi)	-	-	-0.8	0.8	1.6	2.0	-	2.9	3.0	3.7
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Peak Gain (dBi)	4.3	3.2	1.6	1.4	0.3	1.3	-	-	-	-

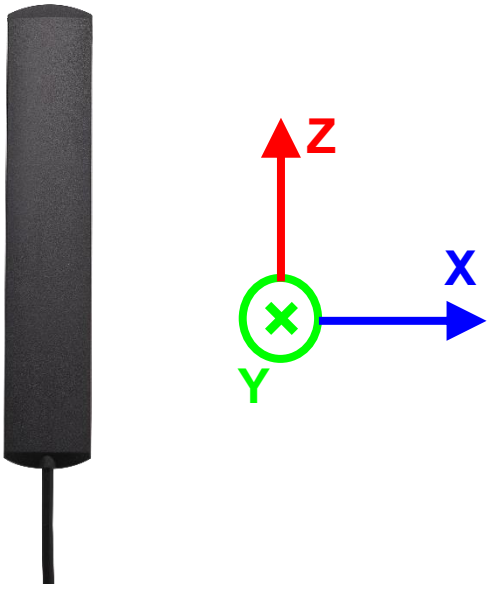
Peak Gain (dBi) – GNSS

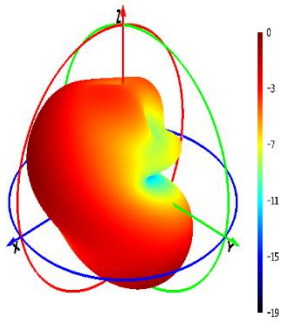
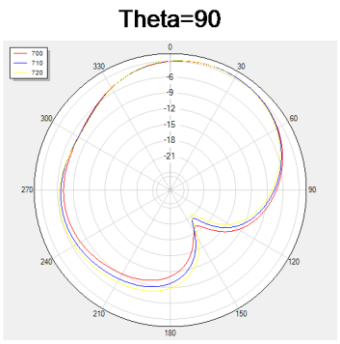
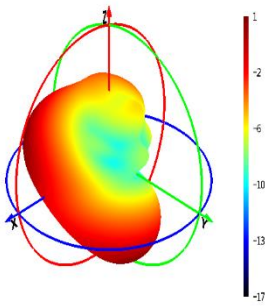
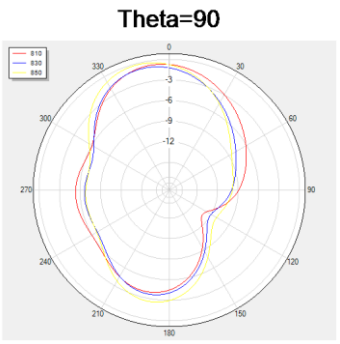
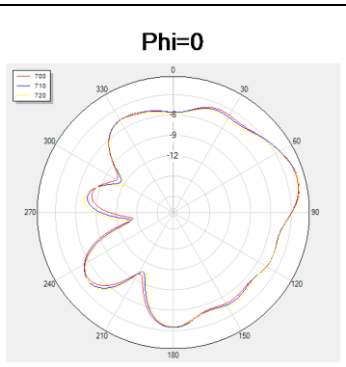
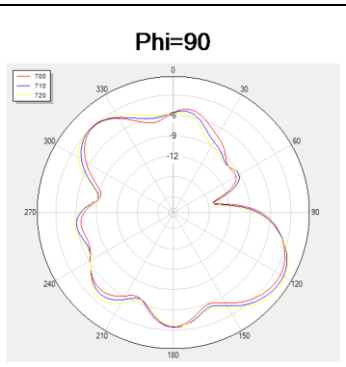
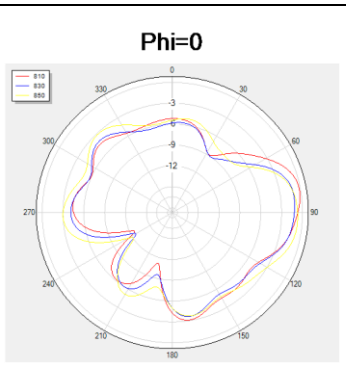
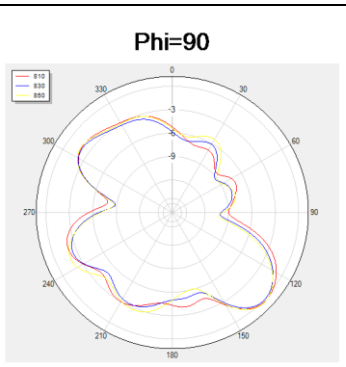
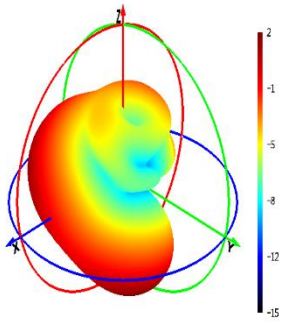
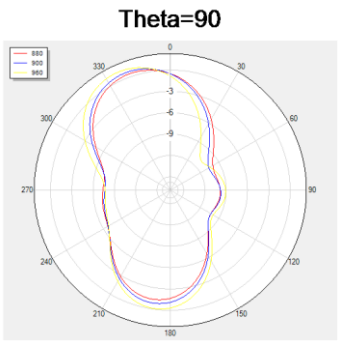
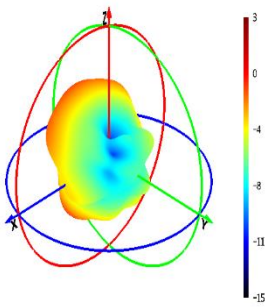
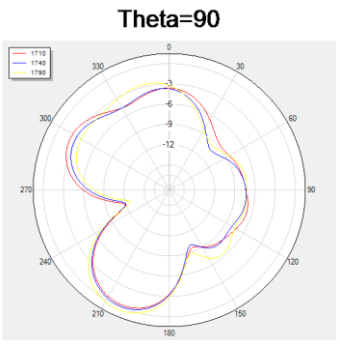
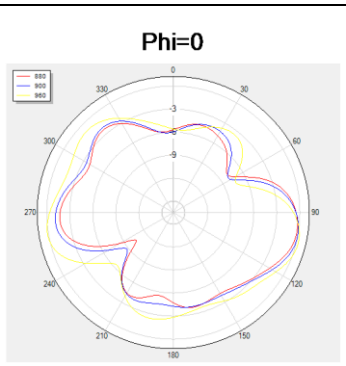
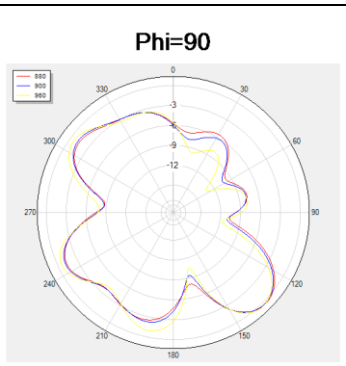
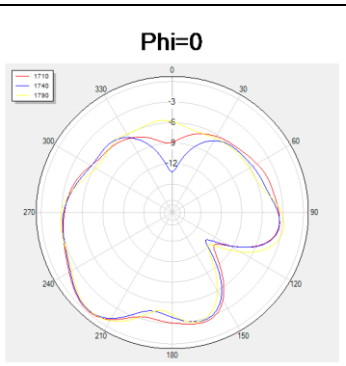
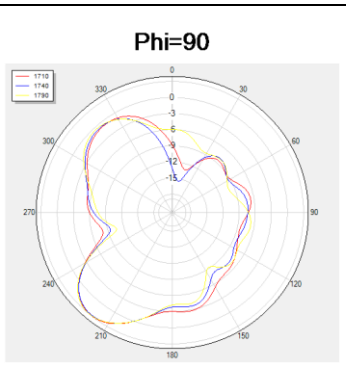
Frequency (MHz)	1176	1207	1227	1248	1268	1561	1575	1602
Peak Gain (dBi)	-	-	-	-	-	-	2.1	-

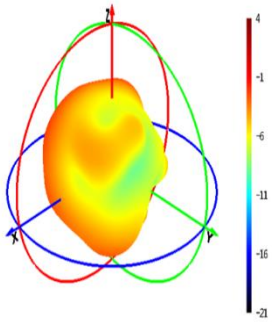
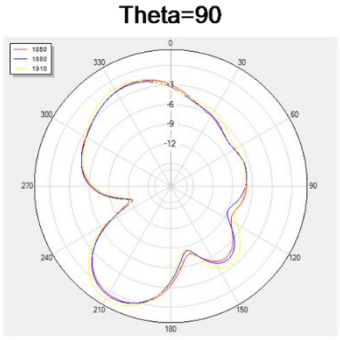
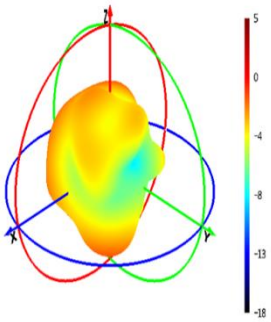
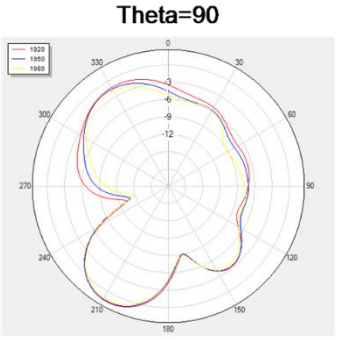
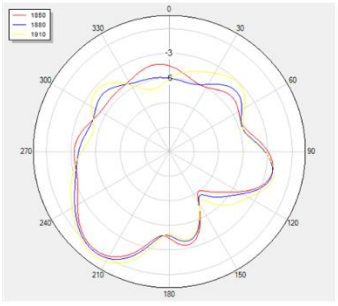
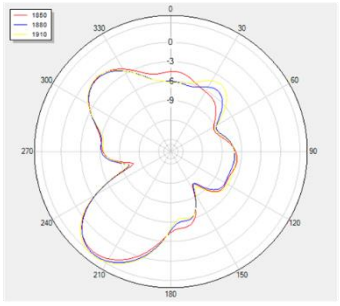
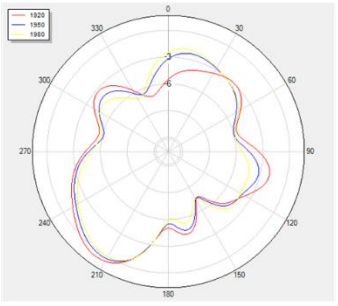
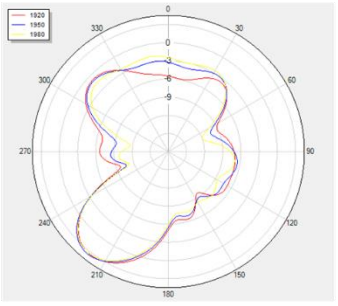
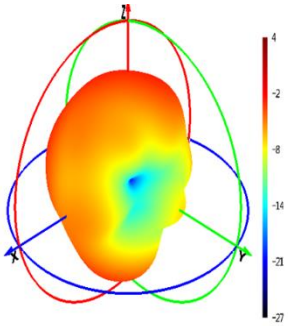
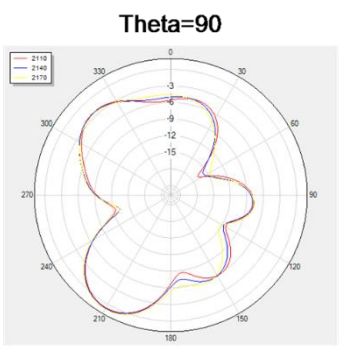
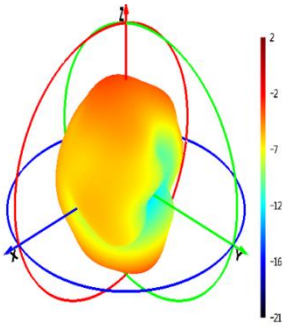
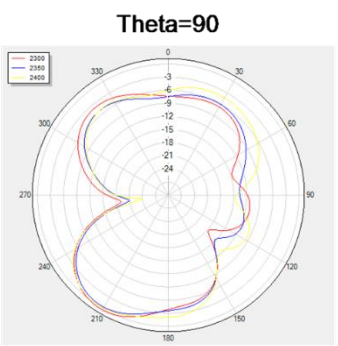
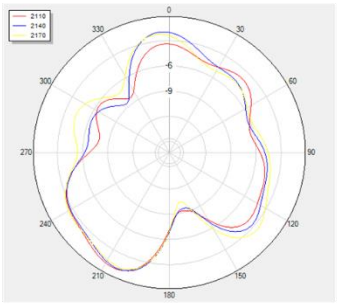
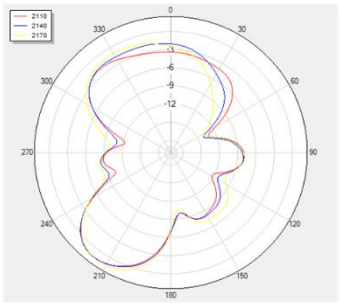
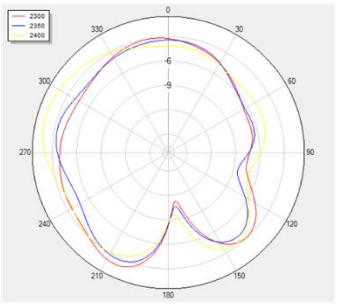
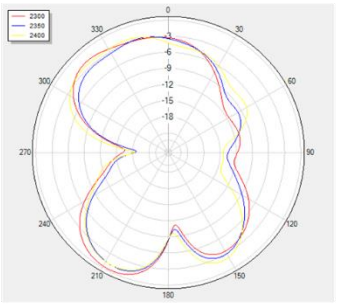
3.2.4. 3D & 2D Radiation Pattern

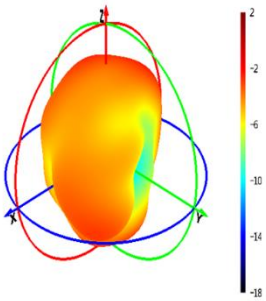
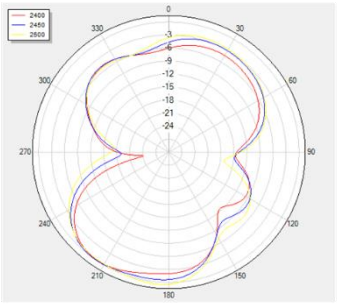
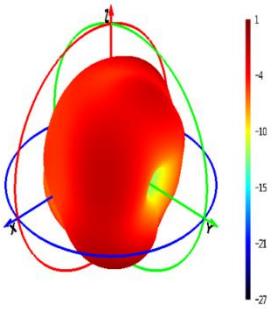
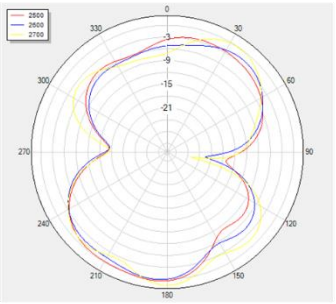
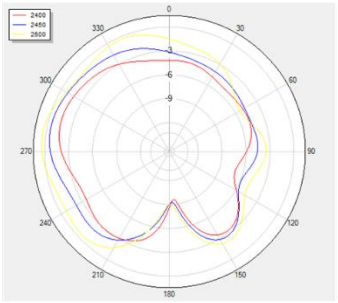
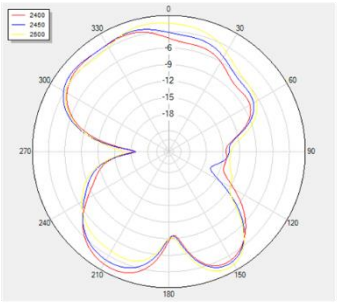
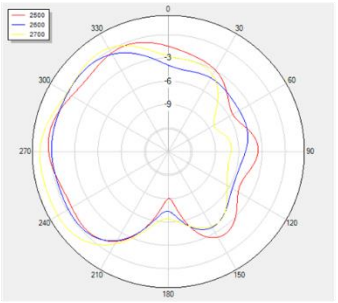
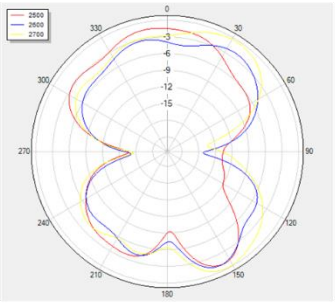
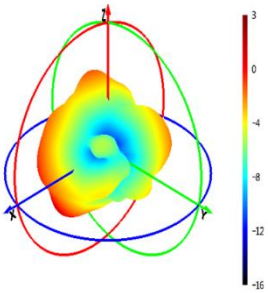
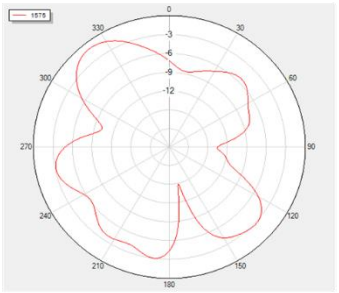
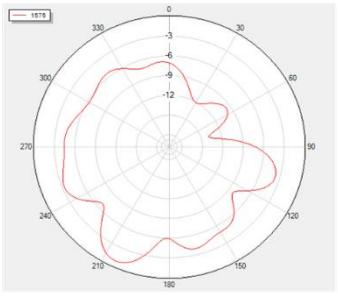
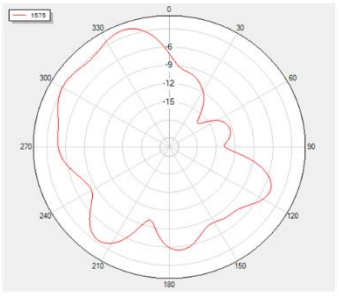
3.2.4.1. Test Condition: Free Space

- Test Chamber: GL-S-1

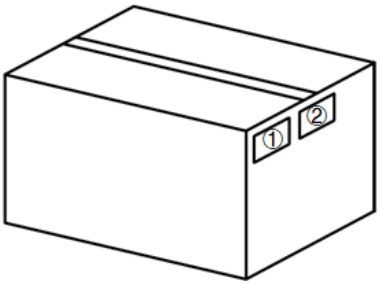


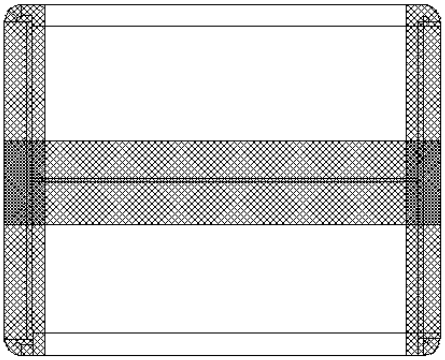
710 MHz		830 MHz	
			
			
900 MHz		1740 MHz	
			
			

1880 MHz		1950 MHz	
			
			
2140 MHz		2350 MHz	
			
			

2450 MHz		2600 MHz	
	Theta=90 		Theta=90 
Phi=0 	Phi=90 	Phi=0 	Phi=90 
1575 MHz			
	Theta=90 		
Phi=0 	Phi=90 		

4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		Single product packaged in small ziplock bag.
2		One large ziplock bag contains 50 products.
3		(7 Large Ziplock Bags / Carton Box) (350 Antennas / Carton Box) Estimated quantity Products that cannot fill the entire carton box are packed in a suitable size carton box. Carton Size: L × W × H = 400 × 290 × 210 mm
4		**Position for Attaching Labels** ① Carton Label ② Quality Label

5		**Sealing Cartons** H-shaped sealing cartons
Note	The initial packaging method described above is for reference only, and the final actual packaging method shall be subject to the actual shipping packaging.	

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

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Tel: +86 21 5108 6236
Email: info@quectel.com

Or our local offices. For more information, please visit:
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Revision History

Version	Date	Author	Note
-	2020-06-30	Kenny Yin	Creation of the document
1.0	2020-06-30	Kenny Yin	First official release
2.0	2020-07-03	Kenny Yin	Updated the specifications (Chapter 3).
2.1	2020-12-16	Kenny Yin	Updated the antenna image (Chapter 2).
2.2	2021-01-27	Kenny Yin	Added IP rating description and installation method (Chapter 3).
3.0	2021-03-29	Kenny Yin/ Aria Chu	Updated all test data in the datasheet and some other data.
3.1	2021-07-05	Kenny Yin	1. Updated product specifications (Chapter 3) 2. Added test condition (Chapter 4).
3.2	2021-07-25	Kenny Yin Aria Chu	1. Updated working temperature (Chapter 3). 2. Added detailed passive electrical specifications (Chapter 3).
3.3	2021-12-01	Kenny Yin/ Aria Chu	Updated the product description (Chapter 1).
4.0	2023-05-11	Joyful Huang/ Lucky Feng/ David Liu/ Aria Chu	Updated all data and datasheet template.
4.1	2023-10-19	Lucky Feng	1. Updated the overview. 2. Updated the mechanical and environmental data (Chapter 1.2). 3. Updated the drawing (Chapter 2).
4.2	2024-02-07	Rhone Wei/ Rainey Liao	Added GPS L1 frequency performance.
4.3	2025-03-06	Allow Xu	Updated the packaging (Chapter 4).
4.4	2025-12-29	Blake Xiang	1. Updated the Overview. 2. Deleted the water resistance rating (Chapter 1.2).

4.5	2026-08-05	Kessie Geng	Updated the packaging (Chapter 4).



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